

Ahmed Safierrahman Abidi

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EDUCATION

ESPRIT School of Engineering

Computer Engineering – EUR-ACE Accredited (5-year integrated diploma)

Grand Tunis, Tunisia

Sep. 2024 – Present

SAMIS High School

High School Diploma – GPA: 4.0 (99.07/100)

Saudi Arabia

Sep. 2021 – Jun. 2024

EXPERIENCE

Software Engineer Intern

Société Tunisienne de l'Electricité et du Gaz (STEG)

Jul. 2025 – Sep. 2025

Ben Arous, Tunisia

- Observed and documented the **SCADA architecture** that supervises Tunisia's national power grid, detailing RTU data acquisition, IEC 60870-5 protocol framing, and real-time database management; the documentation was delivered to the dispatching centre team to support system maintenance.
- Studied **Automatic Generation Control (AGC)** logic and frequency-regulation mechanisms, explaining how the 50Hz grid frequency is kept within a ± 0.2 Hz band; the analysis helped engineers refine control set-points for improved grid stability.
- Examined **redundancy and failover design** across the SCADA stack, including hot-standby server pairs, dual geographic dispatching centres, and RTU backup links via OPGW fibre and microwave; findings were used to update the failover-procedure checklist.
- Reviewed **IT infrastructure** in a critical industrial environment, assessing Red Hat Linux alongside legacy WindowsXP, network segmentation, and OT-specific cybersecurity constraints; recommendations guided the security team's hardening plan.

PROJECTS

Arch Linux Mirror | GNU/Linux (Arch), rsync, systemd, NGINX, Bash, DNS, Git, Docker Dec. 2025 – Present

- Deployed and operate the **first official Arch Linux mirror in Tunisia and North Africa**, serving 70+ GB to 1200+ unique IPs within the first two weeks before any official listing or promotion.
- Automated the full mirror lifecycle using rsync, systemd service/timer orchestration, and fail-fast execution, **reducing manual intervention to zero** across nightly upstream synchronization.
- Built a live metrics dashboard tracking requests, data served, uptime, and all-time unique IPs, accessible at mirror.safiabidi.com/dashboard.
- Navigated 6 months of ISP-level CGNAT constraints to obtain a static public IP and **got accepted for official Arch Linux Tier 2 mirror listing**: <https://archlinux.org/mirrors/safiabidi.com/>

Aquarium Management System | C, PIC16F877, PCB Design, MikroC, Git

Oct. 2025 – Dec. 2025

- Built a PIC16F877-based embedded system monitoring **3 signals** (temperature, water level, lighting) and driving actuators to maintain stable conditions.
- Implemented manual override, real-time status display, and safety alarms using **internal and external interrupts**.
- Persisted thresholds and configuration across power cycles via **EEPROM**.

Starr | PHP, MySQL, HTML/CSS, JavaScript, XAMPP, Git

Oct. 2025 – Dec. 2025

- Developed a **full-stack** educational web platform supporting quizzes, messaging, and safe interaction for children in a team of 6.
- Implemented a points and badges reward system, role-based access control, and content moderation tools.
- Led the design and logic choices for the team to ensure the best quality.

TECHNICAL SKILLS

Programming Languages: C/C++, PHP, JavaScript, HTML/CSS, SQL, Bash

Embedded: PCB Design, MikroC, MPLAB, PIC microcontrollers, interrupt systems, EEPROM

Systems & Infrastructure: Arch Linux, Debian/Ubuntu, server administration, rsync, systemd, NGINX, DNS, Git

Industrial IT Exposure: SCADA architecture, RTU systems, IEC 60870-5, AGC, network segmentation, OT cybersecurity

Languages: English (C2), Arabic (C2), French (B2)